

Exploring 6D Object Pose Estimation with Deformation

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Figure 1. **6D object pose with deformation.** Object rigidity is a core assumption in 6D object pose estimation. While, many objects commonly regarded as rigid can undergo deformation over time due to factors such as collisions, wear from daily use, or improper handling during transport. In this work, we introduce a dataset specifically designed to capture such deformations for 6D object pose estimation. The dataset comprises scanned 3D meshes of 26 everyday objects, each represented in multiple deformed states, with precise mesh alignment across these states. Additionally, we provide 6D pose annotations for these objects in 133K frames, resulting in a total of 665K pose annotations, captured under a variety of conditions. We present the 26 canonical meshes (above), and some examples of captured images (left) and the corresponding scanned meshes (right) showing different levels of deformation.

Abstract

We present DeSOPE, a large-scale dataset for 6DoF deformed objects. Most 6D object pose methods assume rigid or articulated objects, an assumption that fails in practice as objects deviate from their canonical shapes due to wear, impact, or deformation. To model this, we introduce the DeSOPE dataset, which features high-fidelity 3D scans of 26 common object categories, each captured in one canonical state and three deformed configurations, with accurate 3D registration to the canonical mesh. Additionally, it features an RGB-D dataset with 133K frames across diverse

scenarios and 665K pose annotations produced via a semi-automatic pipeline. We begin by annotating 2D masks for each instance, then compute initial poses using an object pose method, refine them through an object-level SLAM system, and finally perform manual verification to produce the final annotations. We evaluate several object pose methods and find that performance drops sharply with increasing deformation, suggesting that robust handling of such deformations is critical for practical applications. The project page and dataset are available at <https://desope-6d.github.io/>.

1. Introduction

Estimating the 6DoF pose of objects is a core task in robotics [24, 26], mixed reality [21, 32, 46], and embodied AI [23, 42, 49]. While existing benchmarks and methods [3, 7, 18, 19, 38] have advanced the field significantly, they largely assume objects are perfectly rigid and match idealized CAD or scanned models—an assumption that rarely holds in practice. Models trained on perfect canonical meshes expect input images to align with these ideal shapes, yet real-world objects often deform unpredictably. In everyday settings, nominally “rigid” items like cardboard boxes, plastic bottles, and metal cans are frequently bent, dented, crushed, or partially collapsed due to regular use or rough handling, as shown in Fig. 1.

Current instance-level datasets and methods [9, 31, 47] offer only limited support for analyzing this regime. They largely focus on intact objects and interpret mesh variation as differences between separate rigid instances. On the other hand, although most category-level pose datasets and methods [6, 25, 27, 28, 36] provide a canonical mesh to represent multiple instances within a category, they lack accurate meshes for individual instances and fail to capture instance-specific geometric variations, as shown in Fig. 2.

We introduce DeSOPE, a real-world RGB-D dataset and 3D asset collection for deformed 6DoF object pose estimation, focusing on everyday items that are nominally rigid but often deformed. For 26 object categories, we capture high-quality scans of one canonical instance and three deformed variants (mild, moderate, severe), aligning all deformed meshes to their canonical counterparts using a flow-driven 3D registration framework [44]. We then collect 133K RGB-D frames across different scenes with these objects. For 6D pose annotation, we label 2D instance masks [34], generate initial poses with an object pose estimator [45], and refine them by jointly optimizing object poses and an implicit neural shape representation [41]. After manual verification, DeSOPE provides 665K high-quality pose annotations across 104 deformed instances.

Using our dataset, we systematically evaluate several typical 6D object pose methods that assume input images align with perfect canonical meshes, which does not hold in this case. Our results show that performance drops significantly as objects deviate from their canonical shapes, highlighting deformation as a major, underexplored limitation in current 6D pose pipelines. To the best of our knowledge, DeSOPE is the first dataset to explicitly capture deformation for 6D object pose estimation.

2. Related Work

Instance-level 6D object pose datasets, including LINEMOD [4], T-LESS [18], YCB-V [7], HOPE [38], and others [1, 14, 17, 35], have been pivotal in advancing object

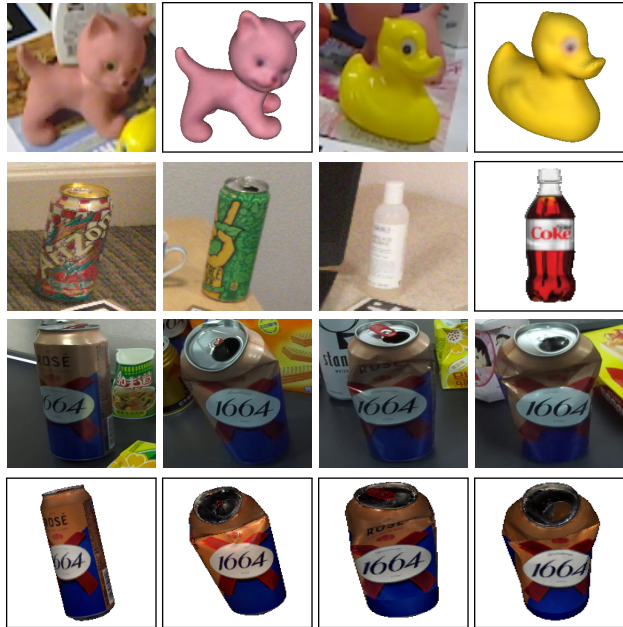


Figure 2. **Comparison of 6D object pose datasets.** The first row shows two examples from an instance-level dataset [4], where each object instance is associated with its own 3D model, under the assumption that the object is perfectly rigid and does not deform over time. The second row depicts a category-level dataset [7], in which multiple instances of the same category share a single 3D model (rightmost). The third row shows the canonical instance from the proposed DeSOPE dataset along with three deformed versions, while the fourth row presents their corresponding 3D meshes. We visualize meshes as black-bordered images on a white background in this figure.

pose estimation. In these datasets, each object instance is paired with its own 3D model—typically a clean CAD geometry or high-quality scan—and is assumed to be perfectly rigid. This one-to-one mapping facilitates precise pose supervision and standardized evaluation across methods.

However, these datasets mostly feature intact, undeformed objects in relatively controlled settings. Even with occlusions, clutter, or hand interactions, each object is assumed to perfectly match its reference model, leaving methods untested against real-world deviations. By contrast, DeSOPE provides physically deformed versions of nominally rigid objects, explicitly aligned to their canonical counterparts, enabling evaluation under realistic deformation.

Category-level object pose datasets aim to model the generalization beyond individual instances by grouping objects into semantic categories [22, 40]. Typically, all instances within a category are represented by a single canonical mesh or template, creating a one-to-many mapping in which diverse objects are assumed to roughly align with the same 3D shape. These datasets are valuable for studying

	Type	Categories	Instances	Images
LINEMOD [4]	Instance	-	15	19K
YCBV [7]	Instance	-	21	80K
PhoCaL [43]	Instance	8	60	3K
Wild6D [14]	Instance	5	162	10K
Objectron [1]	Instance	9	17K	4M
CO3D [35]	Instance	5	19K	1.5M
HANDAL [17]	Instance	17	212	308K
REAL275 [40]	Category	6	42	8K
HouseCat6D [22]	Category	10	194	24K
DeSOPE	Deformed	26	104	133K

Table 1. **Comparison of 6D object pose datasets.** Most instance-level datasets focus on individual instances, assuming objects are perfectly rigid and treating each instance independently. Category-level datasets, on the other hand, provide a single canonical mesh representing all instances within a semantic category. The proposed DeSOPE dataset is the first to capture multiple deformation states of the same instance, offering accurate 3D registration between the deformed state and the canonical form, along with precise 6D object pose annotations in images captured across diverse scenes.

category-level evaluation and for enabling pose estimation when instance-specific CAD models are unavailable.

However, this design limits the study of deformation. Without accurate 3D meshes for each instance, we cannot capture how an object’s pose and geometry change when bent, dented, or compressed in real-world scenarios. By contrast, DeSOPE provides scanned 3D meshes of objects in multiple deformation states, along with precise registration between each deformed mesh and its canonical counterpart. Table 1 summarizes the comparison of different 6D object pose datasets.

Deformable and non-rigid targets have been widely studied in domains such as garments [2, 8, 10, 11], soft bodies [12, 20, 29, 30, 37], and articulated humans [5, 12, 15, 16, 48], where deformation is expected and serves as the primary focus. In contrast, our work targets objects that are nominally rigid—such as packaging and containers—but frequently appear deformed due to damage, wear, or everyday handling. These deviations challenge methods that assume input images match perfect canonical meshes, often causing performance to degrade when objects deform unpredictably in practice. DeSOPE is designed to fill this gap, offering resources that enable the evaluation and future development of deformation-aware 6D object pose estimation techniques.

3. DeSOPE Dataset

This section presents DeSOPE, a real-world dataset for 6D object pose estimation covering diverse object categories and deformable objects. We begin with the 3D scanning

and registration process between canonical and deformed meshes in Section 3.1, and then describe the image acquisition and semi-automatic annotation procedures in Section 3.2. Figure 3 illustrates the overall data collection and annotation framework.

3.1. 3D Model Scanning and Alignment

We select 26 daily object categories. Each category contains one canonical (undeformed) reference instance and three additional instances with progressively increasing deformation levels, categorized as mild, moderate, and severe, resulting in a total of 104 objects, as illustrated in Fig. 3.

We acquire the meshes of all instances using a high-precision Go!SCAN SPARK scanner [13], with an average scanning time of approximately 10 minutes per object.

After acquiring all 3D models, we first obtain an initial registration between the canonical mesh and each deformed mesh through manual alignment. We then refine the alignment using a flow-guided matching strategy. Specifically, we render each mesh from six orthogonal viewpoints (front, back, left, right, top, and bottom), applying the same rotation and translation to ensure consistency across views. To align the deformed mesh with the canonical mesh, we compute dense 2D correspondences between pairs of rendered images from the same viewpoint [44].

Finally, we lift the 2D correspondences to 3D by leveraging the inherent 2D–3D mappings established during rendering, thereby establishing 3D–3D correspondences between the canonical and deformed meshes. For registration, we adopt a two-step strategy: we first apply RANSAC [33] to remove outliers and estimate initial transformation parameters, and then refine the transformation using the Umeyama [39] algorithm, which computes the optimal similarity transformation (including rotation, translation, and scale) over the inlier set, resulting in precise final mesh alignment.

3.2. Image Acquisition and Pose Annotation

Given the reconstructed 3D models, we collect RGB-D video sequences using a ZED2i stereo camera across five common indoor scenes (Fig. 7). For each scene, we capture 208 videos—104 static and 104 dynamic—with five randomly selected object instances placed in the scene. Static videos are recorded by circling around the scene, while dynamic videos involve natural human-object interaction and manipulation.

Accurate pose annotation is essential for dataset utility, yet the annotation process faces two key challenges: (1) handheld camera acquisition introduces motion blur, and objects undergo varying occlusion across viewpoints; (2) stereo depth maps contain sparse noise, and systematic deviations exist between depth-derived point clouds and scanned 3D models. To address these challenges, we pro-

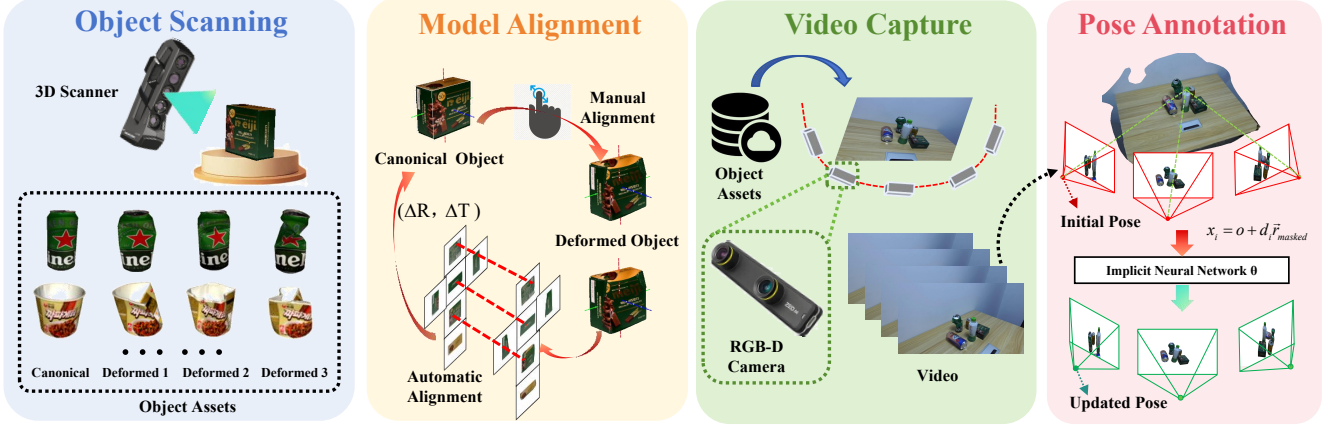


Figure 3. **Overview of the dataset generation framework.** The framework consists of four main steps: **Object Scanning**, which acquires the canonical mesh of objects along with multiple deformed states of the same instance using a high-precision 3D scanner; **Model Alignment**, beginning with coarse manual alignment and followed by flow-driven 3D registration using SCFlow2 [44]; **Video Capture**, which records RGB-D videos of objects across diverse scenes with a stereo camera; and **Pose Annotation**, which performs initial object labeling and iteratively refines poses using implicit neural networks to obtain accurate annotations.

pose a neural network-based annotation framework comprising instance segmentation, initial pose estimation, and pose optimization, as illustrated in Fig. 3 (fourth part).

2D annotation and pose initialization. Given an input RGB-D frame (I_t, D_t) with camera intrinsics $K \in \mathbb{R}^{3 \times 3}$, we first employ a pre-trained segmentation model SAM2 [34] to extract binary masks $\mathcal{M} = \{M_1, \dots, M_5\}$ for $N = 5$ target instances, where $M_i \in \{0, 1\}^{H \times W}$ indicates the i -th instance region. Unlike prior SLAM-based approaches [41] that rely on constant-speed motion models—which struggle under rapid motion and occlusion—we leverage FoundationPose [45] for robust initial pose estimation. FoundationPose estimates the relative pose of each instance via 2D-3D feature matching and PnP solvers, yielding candidate poses $\{\xi_{t,i}^{\text{init}}\}_{i=1}^5$. We perform consistency voting to eliminate outliers, retaining poses with pairwise errors below threshold τ (0.05 rad rotation, 5 cm translation) and computing the averaged initial pose ξ_t^{init} .

Pose refinement. To ensure geometric consistency between camera poses and instance objects, we constrain ray sampling to instance mask regions. Following Co-SLAM [41], we model camera rays as lines from origin o_t with direction $r_{t,u,v}$, but sample only from pixels where $M_i(u, v) = 1$:

$$\mathcal{R}_t = \{(o_t, r_{t,u,v}) \mid \exists i : M_i(u, v) = 1, (u, v) \in \{1, \dots, W\} \times \{1, \dots, H\}\} \quad (1)$$

For each ray, we sample M_c uniform points and M_f depth-guided near-surface points within $[d_{\text{near}}, d_{\text{far}}]$. This mask-constrained sampling ensures that optimization signals derive solely from target instances, reducing background interference.

We jointly optimize camera pose ξ_t and neural scene representation f_θ (mapping world coordinates to color and

TSDF) via a multi-loss objective. Beyond the standard color, depth, SDF, and free-space losses from Co-SLAM, we introduce an instance mask alignment loss $\mathcal{L}_{\text{mask}}$ to enforce pose-instance geometric consistency:

$$\mathcal{L}_{\text{mask}} = \frac{1}{|\mathcal{R}_t|} \sum_{(o_t, r_{t,u,v}) \in \mathcal{R}_t} \left(1 - \max_i M_i(u, v)\right) \cdot \left\| \hat{d}_{t,u,v} - d_{t,u,v} \right\|_2^2 \quad (2)$$

where $\max_i M_i(u, v) = 1$ for instance pixels (no penalty) and 0 for background. The total loss combines all terms with weights $\lambda_{\text{rgb}} = 5$, $\lambda_d = 0.1$, $\lambda_{\text{sdf}} = 1000$, $\lambda_{\text{fs}} = 10$, and $\lambda_{\text{mask}} = 2$:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{rgb}} \mathcal{L}_{\text{rgb}} + \lambda_d \mathcal{L}_d + \lambda_{\text{sdf}} \mathcal{L}_{\text{sdf}} + \lambda_{\text{fs}} \mathcal{L}_{\text{fs}} + \lambda_{\text{mask}} \mathcal{L}_{\text{mask}} \quad (3)$$

We perform global bundle adjustment to jointly optimize all camera poses $\{\xi_t\}_{t=1}^T$ and scene representation f_θ using rays sampled from all keyframe instance masks:

$$\arg \min_{\theta, \{\xi_t\}} \frac{1}{|\mathcal{R}_{\text{global}}|} \sum_{(o_t, r_{t,u,v}) \in \mathcal{R}_{\text{global}}} \mathcal{L}_{\text{total}}(\hat{c}_{t,u,v}, \hat{d}_{t,u,v}, s(x), M_i(u, v)) \quad (4)$$

Our approach differs from Co-SLAM in two aspects: (1) the keyframe ray pool contains only instance mask rays, focusing optimization on object geometry; (2) initial poses derive from FoundationPose rather than constant-speed assumptions, reducing local optima risk. We alternate between optimizing scene parameters θ for $k_m = 10$ iterations

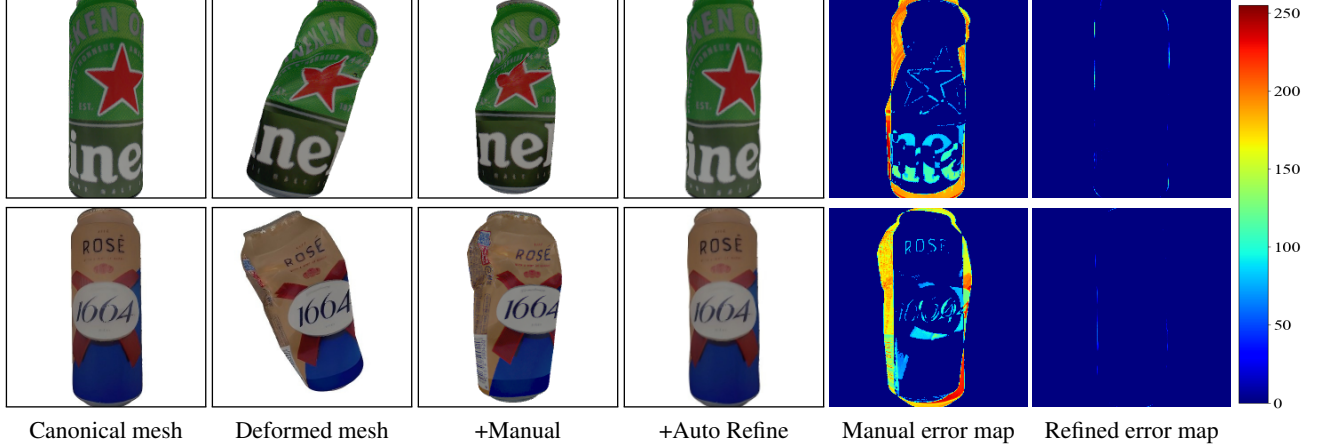


Figure 4. **3D Registration between canonical mesh and deformed mesh.** After the initial manual registration, the two meshes are roughly aligned but still imprecise. Using this manual registration as initialization, the matching-guided automatic refinement step produces a more accurate registration. Note that the reprojection error map after refinement is significantly improved compared to the manual initialization.

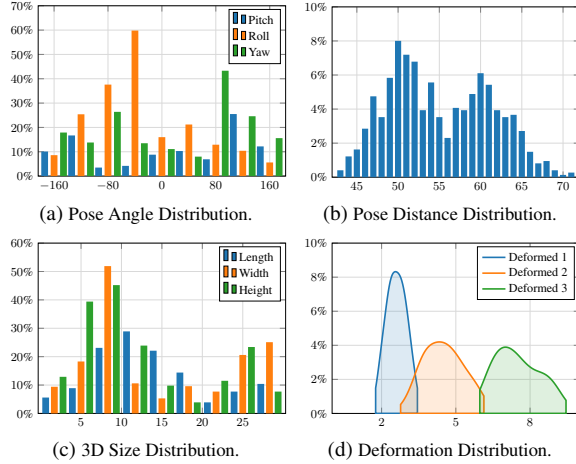


Figure 5. **Statistical analysis of the DeSOPE dataset.** All subplots show percentage (%) on the y-axis. **(a)** Camera pose angle distribution (x-axis: rotation angle in degrees) showing the coverage of pitch, roll, and yaw angles across all annotated frames. **(b)** Object-to-camera distance distribution (x-axis: distance in cm), with distances concentrated around 50cm. **(c)** Physical dimensions of 104 object instances across 26 categories (x-axis: size in cm), demonstrating diversity in object scales. **(d)** Deformation severity distribution (x-axis: deformation magnitude in cm) across three levels: mild (Deformed 1), moderate (Deformed 2), and severe (Deformed 3). Together, these statistics validate the comprehensive coverage of viewing angles, object scales, and deformation states in DeSOPE.

and updating poses using accumulated gradients. This strategy significantly improves instance-to-camera pose consistency compared to SLAM-based approaches prone to background-induced drift.

26 categories of objects. We collect 26 categories of daily

objects, each with four instances: one canonical mesh and three deformed variants of increasing severity, totaling 104 objects, as shown in Fig. 1. The dataset spans common materials and diverse geometries, including both regular and irregular shapes, with broad scale variation. We capture realistic deformation types such as stretching, bending, compression, and twisting, and categorize them into three levels: mild, moderate, and severe, as analyzed in Fig. 5.

5 daily scenarios. We collect the image dataset across multiple indoor scenarios, including a conference room, dining area, window-side setting, sofa area, and break room. For each scene, we capture 208 videos, half of which involve hand-object interactions. We show example images and corresponding pose annotations in Fig. 7.

260 minutes of recordings. We capture the DeSOPE dataset using a ZED2i stereo camera. We record each video for approximately 30 seconds at 30 FPS, yielding 120–240 sampled frames. In total, we collect 133,000 valid RGB-D frames at a resolution of 1920×1080, with each frame containing five object instances, including both deformed and undeformed objects arranged in random combinations with inter-object occlusions. In total, we obtain 665K valid pose annotations.

4. Evaluation

In this section, we first present quantitative and qualitative results of our 3D model alignment procedure in Section 4.1. We then evaluate state-of-the-art 6D object pose estimation methods on DeSOPE across different deformation levels in Section 4.2. Finally, we analyze the factors contributing to performance degradation with increasing deformation severity in Section 4.3.



Figure 6. **Effect of pose refinement.** Contours of projected poses (from FoundationPose and our approach) overlaid on the input image, reflecting the spatial alignment between estimated poses and the target object. Our predicted pose aligns the object more accurately here.



Figure 7. **Example of captured images and pose annotations.** The green boundary contours in the figure represent the results of projecting the poses, obtained by the annotation algorithm proposed in this paper, onto the 2D space. It includes images from both static scenes and dynamic scenarios manipulated by humans.

4.1. Models Alignment Results

As described in Section 3.1, we employ a multi-view point cloud registration method to align the deformed and undeformed 3D models. The core component involves computing the 2D-2D pixel correspondences between the deformed and undeformed 3D models from six different viewpoints.

Since the 3D models are real-world scanned objects without ground-truth 2D-2D correspondences for training, we adopt the large-scale pre-trained optical flow prediction method from SCFlow2 [44] (which jointly estimates optical flow and object pose). Specifically, we utilize approximately 90K publicly available 3D models from datasets including ShapeNet-Objects, Google-Scanned-Objects, and Objaverse to render around 9 million 2D image pairs. These

image pairs are used to train the optical flow network, enabling optical flow estimation for real-world deformed and undeformed 3D model pairs.

The results in Fig.4 demonstrate that the optical flow predictions obtained through our large-scale pre-trained method can effectively match the 2D image pairs of deformed and undeformed 3D models, thereby achieving successful 3D model alignment. Based on the alignment results from each view, we compute quantitative metrics for 3D model alignment, as shown in Table 2. The results indicate that our method can effectively handle 3D model alignment tasks with varying degrees of deformation.

Deformed 1		Deformed 2		Deformed 3	
Init.	+Refine	Init.	+Refine	Init.	+Refine
4.73	2.61	9.38	4.40	11.77	10.10
6.57	1.91	10.50	5.11	13.06	12.53
19.22	11.29	23.99	13.58	12.91	10.91
7.31	2.97	9.67	5.77	9.80	8.43
8.55	4.98	10.98	5.76	6.72	6.90
9.88	6.51	15.30	9.31	10.76	8.78
7.82	5.38	11.38	7.19	14.00	9.33

Table 2. **Alignment Errors of Deformed and Undeformed Models.** "Init." represents the matching error results of manually aligning deformed and undeformed objects, while "+Refine" denotes the matching error results after optimization using SCFlow2. The six middle rows of the table show the alignment error results of the six projected faces respectively, and the last row presents the overall alignment error results.

4.2. Performance of SOTA Algorithms on DeSOPE

Due to varying degrees of object deformation, significant changes occur in both geometric and textural features. To address the challenges in pose estimation on the DeSOPE dataset, we conduct a comprehensive benchmarking of several representative pose estimation algorithms, including category-level methods such as GenPose [50], and unseen object pose estimation methods like SCFlow2 and FoundationPose.

For evaluation metrics, we adopt the standard metrics from the BOP challenge [19], including Visible Surface Discrepancy (VSD), Maximum Symmetry-Aware Surface Distance (MSSD), and Maximum Symmetry-Aware Projection Distance (MSPD). VSD measures the overlap between estimated and ground truth poses on visible surfaces, MSSD evaluates the surface distance over the entire object, and MSPD assesses the projection distance error on the object surface. Following the BOP evaluation protocol, we report the Average Recall (AR), defined as the mean of the recall scores computed from VSD, MSSD, and MSPD, as the primary evaluation criterion.

Regarding data usage, FoundationPose and SCFlow2 are employed without additional training, using their pre-trained models directly. These methods take RGB-D images, instance masks, and undeformed 3D models as input. For GenPose, during training and testing on the DeSOPE dataset, both deformed and undeformed instances in the RGB-D data are matched with their corresponding undeformed 3D models.

As shown in Table 3 and Fig. 8, all three algorithms demonstrate superior pose estimation performance when using original undeformed meshes for inference in static scenes. However, when processing RGB-D data of de-

	SCFlow2	FoundationPose	GenPose
Canonical	0.82	0.78	0.67
Deformed 1	0.67	0.58	0.56
Deformed 2	0.43	0.38	0.36
Deformed 3	0.23	0.24	0.31
Canonical	0.77	0.72	0.61
Deformed 1	0.64	0.54	0.53
Deformed 2	0.34	0.30	0.37
Deformed 3	0.20	0.20	0.28

Table 3. **Benchmark evaluation of state-of-the-art 6D object pose estimation methods on DeSOPE.** We report Average Recall (AR) from the BOP challenge for three representative methods: SCFlow2, FoundationPose (unseen object methods), and GenPose (category-level method). **Rows:** four deformation levels from canonical (undeformed) to increasingly severe deformations. **Columns:** three different pose estimation approaches. **Groups:** static scenes (upper) and dynamic scenes with hand-object interaction (lower). Results show that all methods degrade as deformation increases, with dynamic scenes consistently underperforming static scenes due to occlusions and motion blur.

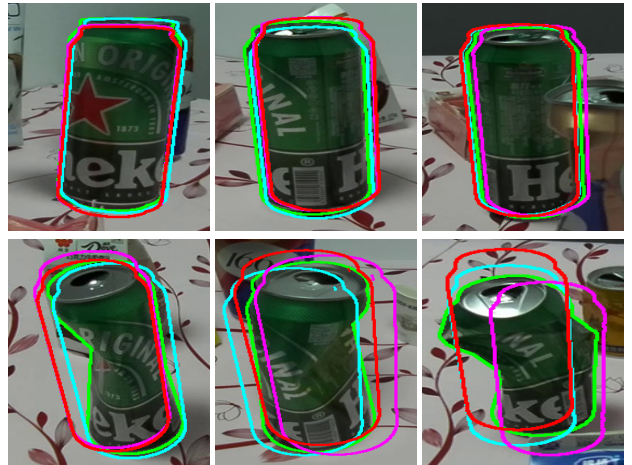


Figure 8. **Evaluating object pose methods on DeSOPE.** The pose estimation results are projected onto the 2D plane using the undeformed canonical mesh applied during inference. Although most state-of-the-art estimators perform well on canonical meshes (the first row), their performance degrades significantly when the meshes exhibit deformations that deviate from the canonical state. Red: GenPose, Pink: FoundationPose, Cyan: SCFlow2.

formed objects, the performance of all methods degrades since the inference relies on undeformed 3D models. Furthermore, the performance deteriorates as the degree of deformation increases. A similar pattern is observed in dynamic scene results. Overall, the pose estimation performance in dynamic scenes is lower than in static scenes, primarily due to hand occlusions.

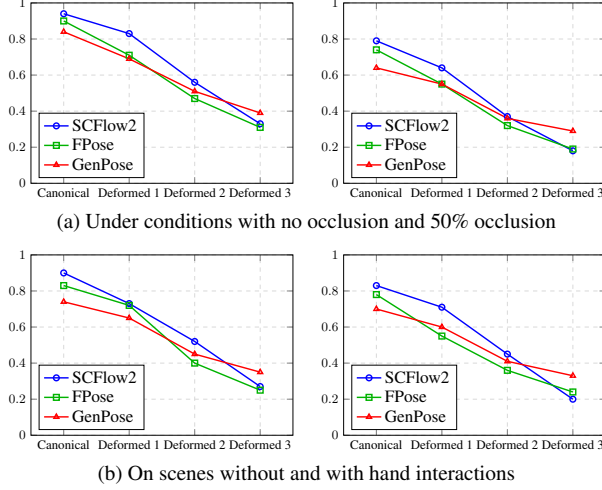


Figure 9. **Performance analysis.** All plots show Average Recall (AR) versus four deformation levels (Canonical to Deformed 3) for three methods: SCFlow2, FoundationPose (FPose), and GenPose. **Top row:** Effect of occlusion—left shows fully visible objects (mask_visib = 1.0), right shows partial occlusion (mask_visib = 0.5). **Bottom row:** Effect of scene type—left shows static scenes, right shows dynamic scenes with hand-object interaction. Key observations: (1) AR decreases as deformation severity increases across all conditions; (2) higher occlusion leads to lower performance; (3) dynamic scenes consistently underperform static scenes due to motion blur and occlusions during manipulation.

4.3. Factors Affecting Performance

We compare method performance on our dataset at different occlusion ratios (AR at occlusion ratio zero and 0.5, respectively). As illustrated in the top row of Fig. 9, greater deformation leads to much poorer performance, and higher occlusion further reduces accuracy.

We further analyze the performance disparity between static and dynamic scenarios. As illustrated in the bottom row of Fig. 9, all three algorithms consistently achieve higher accuracy in static scenes compared to dynamic scenes across all deformation levels. This performance gap can be attributed to two primary factors: (1) hand occlusions during object manipulation significantly reduce the visible surface area, limiting the geometric features available for pose estimation; and (2) motion blur introduced by rapid hand movements degrades the quality of RGB-D observations, affecting both feature extraction and depth estimation. Notably, the performance degradation from static to dynamic scenes is more pronounced for methods that rely heavily on precise geometric matching, such as SCFlow2 and FoundationPose, while GenPose exhibits relatively smaller gaps due to its category-level generalization capability.

5. Conclusion

We introduced DeSOPE, the first dataset designed to study 6D object pose estimation under realistic deformation, addressing a critical gap in existing benchmarks. Through comprehensive evaluation, we demonstrate that current state-of-the-art methods suffer substantial performance degradation as deformation increases, revealing a fundamental limitation of rigid-object assumptions. By providing aligned 3D meshes across deformation states and large-scale pose annotations, DeSOPE establishes a challenging and realistic benchmark. We hope this dataset will encourage future research on deformation-aware representations, temporal modeling, and more robust pose estimation methods for real-world applications.

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